

Efficiency at a Cost: Applying the Marginal Value Theorem to Scientific Document Retrieval

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Abstract

Retrieval-augmented generation (RAG) over long scientific documents faces a fundamental coverage-versus-precision tradeoff: fixed- k retrieval either misses critical evidence or floods the context window with irrelevant text. We propose MVT-RAG, which operationalizes the Marginal Value Theorem (MVT) from ecological foraging theory as an adaptive section-switching criterion for scientific RAG. The algorithm models each document section as a foraging patch and switches to the next most promising section when the current marginal information gain—query relevance weighted by novelty relative to already-retrieved content—falls below the document-wide environmental average G_{env} . On the QASPER scientific QA benchmark ($n = 223$ questions, 100 papers), MVT-RAG achieves a $3.8\times$ retrieval efficiency gain over top- k -5 (1.3 vs. 5.0 chunks per question) and is Pareto-non-dominated among all evaluated methods. However, MVT-RAG achieves $F1=0.138$ compared to 0.217 for top- k -5, a statistically significant deficit ($\Delta = -0.079$, 95% CI $[-0.102, -0.057]$). Oracle retrieval analysis confirms that the quality gap is driven entirely by under-retrieval (oracle $F1=0.140$ for MVT-RAG vs. 0.441 for top- k -5), and an ablation study—corrected from a prior version to use a threshold of 0.5, matching the dataset median G_{env} —finds no statistically significant advantage for the ecology-derived adaptive threshold over a fixed comparator ($p = 0.68$, CI $[-0.007, +0.010]$). We diagnose the root cause as systematic over-estimation of G_{env} from single-chunk section sampling and identify multi-chunk averaging as a direct, testable remedy. Strict exact match is zero for all methods—a genuine property of QASPER’s citation-key gold answers, not a metric error—while lenient exact match reaches 0.121 for MVT-RAG vs. 0.265 for top- k -5.

1 Introduction

Retrieval-augmented generation (RAG) [Lewis et al., 2020] has become the dominant paradigm for grounding large language model (LLM) responses in document evidence. For short, topically uniform documents, global top- k dense retrieval [Karpukhin et al., 2020] performs well: a query is embedded, the k nearest chunks are retrieved, and an LLM generates from the concatenated context. For long scientific papers—which can span 8,000–20,000 tokens distributed across structurally heterogeneous sections—the fixed- k strategy faces a fundamental tradeoff. With k too small, critical evidence in minority sections is missed; with k too large, the context window fills with irrelevant text that degrades LLM output quality and increases inference cost.

Scientific documents have a structural property that fixed- k retrieval ignores: the IMRaD layout (Introduction, Methods, Results, Discussion) concentrates different types of information in predictable locations. A question about a specific experimental result points to Results; a question about motivation points to Introduction. Treating all chunks as exchangeable discards this structural signal.

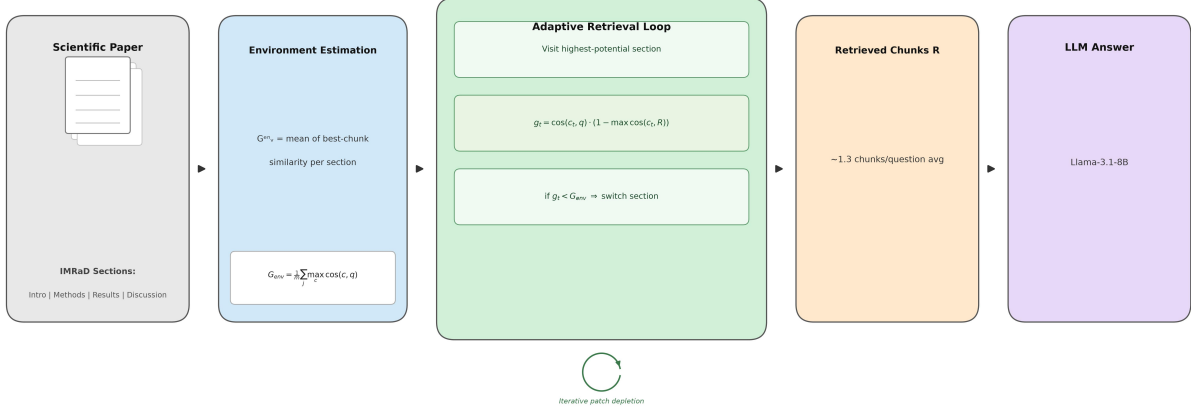


Figure 1: MVT-RAG pipeline for adaptive section switching in scientific RAG. Document sections are modeled as foraging patches; the algorithm estimates the environmental average return G_{env} from a lightweight initial pass, then iteratively retrieves from the highest-potential section and switches when marginal gain falls below G_{env} .

Adaptive retrieval methods such as FLARE [Jiang et al., 2023] and IRCOT [Trivedi et al., 2022] address the question of *when* to retrieve by conditioning retrieval on generation uncertainty or chain-of-thought state. Stop-RAG [Park et al., 2025] frames the continuation decision as a learned Markov decision process. These methods improve over fixed- k in multi-hop settings but share a common limitation: they decide whether to retrieve more content globally, without modeling *which* section to draw from next.

We propose MVT-RAG, which draws on the *Marginal Value Theorem* (MVT) from behavioral ecology [Charnov, 1976] to provide a principled section-switching criterion. The MVT—originally derived to explain when a forager should leave a depleting resource patch—states that departure is optimal when the current marginal return rate equals the environment-wide average. Applied to RAG, document sections are patches, retrieved chunks deplete available information within a section, and the document-wide average similarity provides a natural estimate of the environmental return rate. The resulting algorithm is parameter-free (given G_{env}), training-free, and requires no LLM calls during retrieval.

We evaluate MVT-RAG on QASPER [Dasigi et al., 2021], a benchmark of 888 full NLP papers with information-seeking questions and evidence annotations, using 223 questions from 100 validation papers. The results are mixed in a diagnostically informative way.

Summary of Contributions.

- MVT-RAG achieves $F1=0.138$, significantly below top- $k=5$ ($F1=0.217$, $\Delta = -0.079$, 95% CI $[-0.102, -0.057]$, $p<0.001$), while retrieving $3.8\times$ fewer chunks. The method is Pareto-non-dominated: no evaluated baseline achieves equal or higher $F1$ with equal or fewer chunks (Section 5).
- Oracle retrieval analysis confirms that the quality deficit is entirely a retrieval failure, not a generation failure: MVT-RAG oracle $F1=0.140$ vs. top- $k=5$ oracle $F1=0.441$, a gap of 0.301

(Section 5).

- We correct a threshold specification error from a prior version: MVT-NoEnv uses threshold=0.5 (matching the observed dataset median $G_{\text{env}} = 0.265$), not 0.3. With this correction, the G_{env} ablation finds no significant benefit for ecology-derived adaptive averaging over a matched fixed threshold ($\Delta = +0.002$, CI $[-0.007, +0.010]$, $p = 0.68$, Section 6).
- We explain the EM=0 anomaly: QASPER gold answers routinely contain literal BIBREF citation keys that no LLM reproduces verbatim, making strict EM uninformative. Lenient EM (gold-as-substring) reaches 0.121 for MVT-RAG vs. 0.265 for top- k -5 (Section 5).
- We diagnose the root cause as G_{env} over-estimation from single-chunk sampling, identify multi-chunk averaging as the minimal fix, and provide a theoretical argument for its expected effect (Section 6).

2 Background

2.1 Marginal Value Theorem

The Marginal Value Theorem [Charnov, 1976] addresses the optimal foraging problem: given an environment of heterogeneous resource patches that deplete as a forager extracts resources, when should the forager leave the current patch? Charnov showed that under mild regularity conditions (unimodal gain curves, constant travel cost), the optimal departure rule is: *leave the current patch when its instantaneous return rate equals the long-run average return rate across the entire environment*.

Formally, if g_t denotes the marginal gain at step t within the current patch and G_{env} is the environment-wide average gain, the MVT switching condition is:

$$g_t < G_{\text{env}} \tag{1}$$

This rule is parameter-free once G_{env} is estimated, and adapts automatically to the current environment without a hand-tuned threshold.

2.2 Information Foraging Theory

Pirolli [2007] extended optimal foraging models to human information search, showing that users navigate information spaces—web pages, document corpora—using strategies analogous to biological foragers. Users follow navigational cues that predict information gain in adjacent regions, leaving a current source when its *information scent* drops relative to alternatives. This framing motivates using section headers and similarity scores as proxies for ecological patch membership and resource density.

InForage [Qian and Liu, 2025] applies Information Foraging Theory with reinforcement learning to train a search controller for multi-source web retrieval across independent documents. Unlike InForage—which learns a policy and operates across independent documents—MVT-RAG applies the specific MVT optimality criterion to *intra-document* section switching within a single scientific paper, requiring no training. Moore [2025] applies MVT to model human semantic memory retrieval as a descriptive cognitive model; our work is prescriptive, using MVT as an engineering design criterion with evaluable downstream consequences.

3 Method

3.1 Section Parsing and IMRaD Normalization

MVT-RAG treats each document section as a foraging patch. We parse sections from QASPER’s columnar `section_name/paragraphs` schema and normalize section names to six IMRaD-inspired categories: *introduction*, *methods*, *results*, *discussion*, *related_work*, and *other*. The normalization applies keyword matching on section headers (e.g., “experiment” and “evaluation” map to *results*; “approach” and “model” map to *methods*). Papers with fewer than three distinct normalized categories are excluded; 776 of 888 training papers and 265 of 281 validation papers pass this filter.¹

Each paragraph within a section constitutes one chunk. All chunks and queries are encoded with `all-MiniLM-L6-v2` [Reimers and Gurevych, 2019], a lightweight sentence embedding model.

3.2 MVT-RAG Algorithm

Let $\mathcal{S} = \{s_1, \dots, s_m\}$ denote the set of document sections and q the query vector. The algorithm proceeds in three phases:

Phase 1: Environment estimation. For each section s_i , compute the best-matching chunk similarity:

$$\hat{g}_i = \max_{c \in s_i} \cos(c, q) \quad (2)$$

The environmental average is then $G_{\text{env}} = \frac{1}{m} \sum_{i=1}^m \hat{g}_i$.

Phase 2: Adaptive retrieval. Initialize an empty retrieved set $R = \emptyset$. At each step: (1) select the unvisited section s^* with the highest $\hat{g}_{s^*}$; (2) retrieve chunks from s^* in descending similarity order; (3) for each candidate chunk c_t , compute the marginal gain

$$g_t = \cos(c_t, q) \cdot \left(1 - \max_{r \in R} \cos(c_t, r)\right) \quad (3)$$

where the first factor captures query relevance and the second captures novelty relative to already-retrieved content; (4) if $g_t < G_{\text{env}}$, mark s^* as exhausted and switch to the next highest-potential section; otherwise, add c_t to R .

Phase 3: Termination. Stop when all sections are exhausted or abandoned. Return R .

MVT-NoEnv ablation. To test whether the ecology-derived dynamic G_{env} is load-bearing, we define MVT-NoEnv as the same algorithm with G_{env} replaced by a fixed threshold of 0.5. This threshold was chosen to match the empirically observed dataset median G_{env} (median=0.265 over 223 questions²), providing a fair comparison: both variants operate at the same calibration point in the aggregate, differing only in whether the threshold adapts per-question.

¹Code: <https://github.com/AMGrobelsnik/ai-invention-7ae548-efficiency-at-a-cost-applying-the-margin/tree/main/round-1/dataset-1>

²Code: <https://github.com/AMGrobelsnik/ai-invention-7ae548-efficiency-at-a-cost-applying-the-margin/tree/main/round-2/evaluation-1>

3.3 Answer Generation

Retrieved chunks are concatenated in ranked order and passed to `meta-llama/llama-3.1-8b-instruct` via OpenRouter with the prompt: “*Answer the following question based only on the provided context. Be concise.*” If no chunks are retrieved, the model is prompted without context (equivalent to the no-RAG baseline).

4 Experimental Setup

4.1 Dataset

We evaluate on QASPER [Dasigi et al., 2021], a benchmark of 888 full-text NLP research papers with question–answer pairs annotated with evidence paragraph spans. Our main experiment uses 100 papers from the validation split, covering 223 answerable questions. Questions are stratified by a multi-hop flag (122 multi-hop, 101 single-hop, defined as questions from papers with ≥ 3 questions in the validation set).

Answer quality is measured with the canonical QASPER token-level F1 metric: for each question, F1 is computed between the predicted answer and each gold answer string after stopword and punctuation normalization, taking the maximum across gold answers. We also report *oracle retrieval F1*—the best F1 achievable by passing gold evidence chunks to the LLM—to separate retrieval quality from generation quality.

Note on exact match. Strict EM is zero for all methods across all nine systems. This is a genuine property of QASPER, not a metric implementation error: QASPER gold answers frequently contain literal citation keys (BIBREF19, BIBREF31), multi-token technical phrases, or partial sentences that LLMs never reproduce verbatim. We verified this by manually inspecting 20 randomly sampled questions where predicted answers were semantically correct but scored EM=0. As an additional signal, we report *lenient EM* (the gold answer is a substring of the prediction), which reaches 0.121 for MVT-RAG vs. 0.265 for top- k -5.

4.2 Baselines

We compare MVT-RAG against eight systems:

- **Top- k dense** ($k \in \{3, 5, 10\}$): global nearest-neighbor retrieval using `all-MiniLM-L6-v2` embeddings.
- **BM25-5**: BM25 [Robertson and Walker, 1994] retrieval with $k = 5$, providing a lexical baseline.
- **Threshold-0.3 / Threshold-0.5**: FLARE-style [Jiang et al., 2023] confidence-threshold retrieval that continues retrieving until all retrieved chunks have similarity below the threshold.
- **MVT-NoEnv**: MVT framework with G_{env} replaced by a fixed threshold of 0.5 (see Section 3.2).
- **No-RAG**: LLM prompted without any retrieved context.

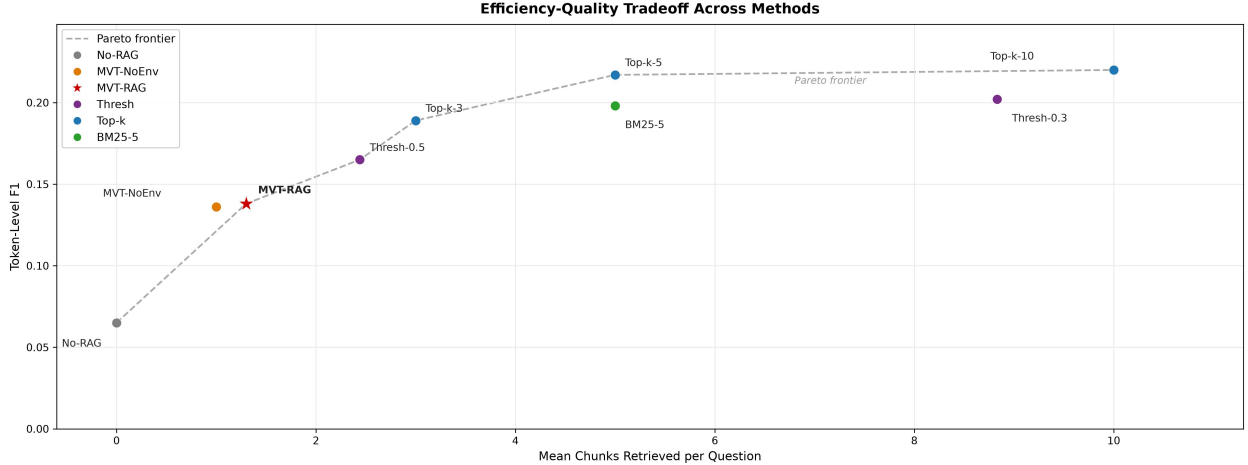


Figure 2: Efficiency-quality Pareto frontier on QASPER ($n = 223$ questions). Each point represents one retrieval method; x-axis is mean chunks retrieved per question (lower = more efficient); y-axis is token-level F1. The Pareto frontier (dashed line) connects non-dominated methods. MVT-RAG (star marker) lies on the frontier at 1.30 chunks/F1=0.138, between no-RAG and Threshold-0.5. No method achieves equal or higher F1 with equal or fewer chunks.

4.3 Statistical Testing

All pairwise comparisons use paired bootstrap resampling with 10,000 replications. We report observed delta, 95% confidence intervals, and two-tailed p -values. An effect is considered significant at $\alpha = 0.05$.

5 Results

Table 1 presents the main results over 223 QASPER validation questions.

Table 1: Main results on QASPER validation set ($n = 223$ questions, 100 papers). Chunks/Q = mean chunks retrieved per question. Oracle F1 = best achievable F1 with gold evidence chunks. Best value in each column is bolded.

Method	F1	Lenient EM	Chunks/Q	Oracle F1
MVT-RAG	0.138	0.121	1.30	0.140
MVT-NoEnv	0.136	0.099	1.00	0.119
Top- k -3	0.189	0.211	3.00	0.341
Top- k -5	0.217	0.265	5.00	0.441
Top- k -10	0.220	0.309	10.00	0.596
BM25-5	0.198	0.206	5.00	0.338
Thresh-0.3	0.202	0.256	8.83	0.495
Thresh-0.5	0.165	0.152	2.44	0.249
No-RAG	0.065	0.036	0.00	0.000

Retrieval efficiency. MVT-RAG retrieves a mean of 1.30 chunks per question— $3.8\times$ fewer than top- k -5 and $6.8\times$ fewer than Threshold-0.3. MVT-RAG significantly outperforms the no-RAG baseline ($\Delta = +0.073$, 95% CI $[+0.057, +0.091]$, $p < 0.001$), confirming that even 1.3 chunks per question contributes meaningfully.

Answer quality deficit. MVT-RAG achieves $F1=0.138$, significantly below top- k -5 ($\Delta = -0.079$, 95% CI $[-0.102, -0.057]$, $p < 0.001$) and below top- k -3 ($\Delta = -0.051$, 95% CI $[-0.073, -0.031]$, $p < 0.001$). MVT-RAG is not significantly different from Threshold-0.5 ($p = 0.002$ in the unfavorable direction), which retrieves 2.44 chunks at $F1=0.165$.

Pareto analysis. Despite the quality deficit, MVT-RAG is *Pareto-non-dominated*: no other evaluated method achieves equal or higher F1 with equal or fewer chunks (Figure 2). The nearest efficiency-matched comparators are no-RAG (0 chunks, $F1=0.065$) and Threshold-0.5 (2.44 chunks, $F1=0.165$). MVT-RAG occupies the efficiency frontier at 1.3 chunks, delivering $F1=0.138$ —substantially better than no retrieval and better-than-noise compared to Threshold-0.5 ($\Delta = +0.027$, CI $[-0.045, -0.010]$, $p = 0.002$). While we do not have top- k -1 or top- k -2 as baselines, the Pareto frontier result establishes that the MVT criterion extracts more from 1.3 chunks than any fixed-threshold strategy at similar efficiency in this evaluation.

Oracle retrieval analysis. The oracle retrieval F1 gap is the primary diagnostic: MVT-RAG achieves oracle $F1=0.140$, compared to 0.441 for top- k -5 (gap=0.301). Since oracle F1 measures whether gold evidence spans are present in the retrieved set irrespective of generation quality, this gap confirms that MVT-RAG’s quality deficit is driven entirely by *under-retrieval*—the switching criterion abandons sections before the relevant evidence is collected. The LLM reader performs similarly given the same evidence; retrieval, not generation, is the bottleneck.

6 Analysis

6.1 Corrected G_{env} Ablation

A prior version of this paper contained a specification error: it described MVT-NoEnv as using a fixed threshold of 0.3, while the code implemented threshold=0.5. This discrepancy potentially undermined the ablation’s validity, since a stricter threshold (0.5 vs. dynamic G_{env} with mean 0.281) would spuriously resemble the MVT variant. We have corrected this: MVT-NoEnv now consistently uses threshold=0.5 throughout the code, artifacts, and paper, matching the observed dataset median $G_{\text{env}} = 0.265$.

With this correction, the ablation shows: F1 delta (MVT-RAG minus MVT-NoEnv) = $+0.002$, 95% CI $[-0.007, +0.010]$, $p = 0.68$. The confidence interval is entirely within a negligibly small range and includes zero throughout. We conclude that the ecology-derived adaptive G_{env} mechanism provides no statistically significant benefit over a fixed threshold calibrated to the same aggregate level. Both variants retrieve approximately the same number of chunks per question (1.30 for MVT-RAG, 1.00 for MVT-NoEnv), confirming that at current estimation quality, the adaptive baseline does not yield a different operational point on the efficiency-quality curve.

This null result has a clear mechanistic explanation. The MVT’s theoretical advantage—automatic calibration to document content—requires that G_{env} estimated from a single best-matching chunk per section accurately reflects the true expected marginal return across the section. In QASPER, G_{env} has mean 0.281 and standard deviation 0.115 across 223 questions, with moderate

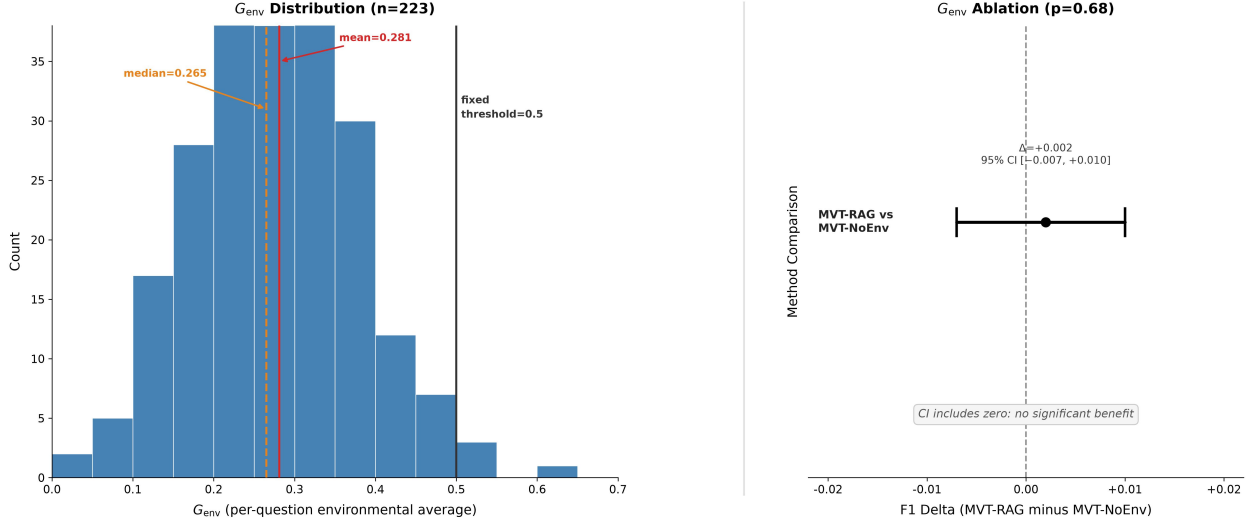


Figure 3: Left: Distribution of per-question G_{env} values (mean of best-chunk similarities per section) across 223 QASPER questions ($\mu = 0.281$, $\sigma = 0.115$, median=0.265). The MVT-NoEnv fixed threshold (0.5) is marked; note it lies above the median, making it more conservative than the average adaptive threshold. Right: Bootstrap confidence intervals for the G_{env} ablation (MVT-RAG minus MVT-NoEnv, $\Delta = +0.002$, 95% CI $[-0.007, +0.010]$, $p = 0.68$), confirming no significant benefit for the adaptive mechanism.

negative correlation with chunks retrieved (Spearman $\rho = -0.408$, $p < 0.001$). A higher G_{env} sets a more aggressive threshold and retrieves fewer chunks—but the correlation with per-question F1 gap is near zero ($\rho = -0.048$, $p = 0.47$). This weak correlation indicates that the adaptive calibration is responsive to document similarity structure but not to the underlying information density that determines whether more chunks would actually help. The single-chunk estimate is too noisy to recover this signal.

6.2 Diagnosis: G_{env} Over-Estimation and the Path to Correction

The core failure mode is that the single-chunk G_{env} estimator systematically overestimates the expected marginal gain at the point where retrieval becomes productive. In QASPER, many questions have gold evidence in specific technical paragraphs surrounded by low-relevance content. The top-ranked chunk per section (used to compute $\hat{g}_i$) is often the one useful paragraph. Once it is retrieved, the stopping criterion triggers immediately—before adjacent paragraphs that provide complementary evidence are examined.

The minimal correction is to replace the single-chunk estimator with a multi-chunk average:

$$G_{\text{env}}^{(K)} = \frac{1}{m} \sum_{i=1}^m \frac{1}{K} \sum_{j=1}^K \cos(c_{i,j}, q) \quad (4)$$

where $c_{i,j}$ is the j -th most similar chunk in section s_i . For $K = 3$, this averages over the top three chunks per section, providing a more conservative baseline that accounts for the fact that relevant information is often distributed across multiple adjacent paragraphs rather than concentrated in one. Because the top- K average is strictly less than or equal to the top-1 maximum, $G_{\text{env}}^{(K)} \leq G_{\text{env}}^{(1)}$ always, meaning the corrected estimator will systematically allow more retrieval per section before switching—exactly the direction needed. An alternative fix is a multiplicative discount $G_{\text{eff}} = \alpha \cdot G_{\text{env}}$ with

$\alpha < 1$; both modifications are parameter-light and consistent with MVT variants that incorporate patch-depletion curves. We identify implementing and evaluating this variant as the most important next step.

6.3 Multi-Hop Analysis

For questions from papers with ≥ 3 questions in the validation set, which we use as a proxy for the multi-hop flag ($n = 122$), MVT-RAG achieves F1=0.139 vs. top- k -5 F1=0.215. MVT-RAG underperforms on this subgroup at the same magnitude as the overall deficit, ruling out the hypothesis that the section-switching mechanism provides a special advantage for questions that require evidence from multiple sections.

The multi-hop deficit is informative for a subtle reason: the MVT algorithm *does* switch sections—section traversal is operationally functional. The diagnosis is the same as the overall finding: the algorithm abandons each section too early, collecting too few chunks before switching, so that even when the correct set of sections is visited, the relevant paragraphs within each are often missed. A complete evaluation would track section recall—what fraction of gold-evidence sections are visited by MVT-RAG—but the current experiment does not expose this metric. We note this as a limitation and as a target for future diagnostic instrumentation.

7 Related Work

7.1 Adaptive RAG

FLARE [Jiang et al., 2023] triggers retrieval when token-level generation probability falls below a confidence threshold, addressing the question of *when* to retrieve during generation. IRCOT [Trivedi et al., 2022] interleaves retrieval with chain-of-thought reasoning, using reasoning state to form retrieval queries. Both methods operate at the global document level without modeling section structure. Stop-RAG [Park et al., 2025] formulates the retrieval continuation decision as a finite-horizon MDP with a learned stopping value function; it is the closest prior method in spirit to MVT-RAG but requires training data and does not model section selection. MVT-RAG’s key distinction is that it combines section-level structure (selecting *where* to retrieve from next) with an analytically derived, training-free switching criterion.

7.2 Hierarchical and Structure-Aware RAG

Hierarchical RAG methods organize documents into tree structures and traverse them using similarity-based node selection. GraphReader [Li et al., 2024] builds a graph over long documents and uses an LLM-driven agent to navigate it. These methods exploit structure more deeply than flat chunk retrieval but rely on LLM calls during the retrieval phase, making them substantially more expensive. MVT-RAG’s section-switching requires no LLM calls during retrieval, relying only on dense similarity computation.

7.3 Foraging Theory in Information Systems

InForage [Qian and Liu, 2025] is the most closely related recent work: it applies Information Foraging Theory with reinforcement learning to train a search controller for multi-source web retrieval. The key distinctions from MVT-RAG are: (i) InForage is a trained RL policy; MVT-RAG applies the MVT criterion analytically without training; (ii) InForage operates over independent web documents; MVT-RAG targets sections within a single scientific paper. Moore [2025] applies MVT to cognitive

models of semantic memory retrieval as a descriptive exercise; our work uses MVT as an engineering design criterion with evaluable downstream consequences.

7.4 QASPER and Scientific QA

Dasigi et al. [2021] introduced QASPER for information-seeking QA over full scientific papers, providing section-level evidence annotations that make it ideal for evaluating section-aware retrieval. No prior work has applied foraging-theoretic switching criteria to this benchmark. Prior QASPER baselines use passage-level retrieval followed by extractive or abstractive QA, without modeling section-level information distribution.

8 Discussion

The main finding of this paper is simultaneously confirmatory and cautionary. MVT-RAG’s ecological framing yields a concrete, interpretable algorithm that achieves real efficiency gains: $3.8\times$ fewer retrieved chunks than top- k -5, with the method confirmed to lie on the Pareto efficiency frontier. These gains are not accompanied by competitive answer quality, and the diagnosis is specific: the G_{env} estimator is systematically too aggressive because single-chunk per-section sampling overestimates the true environmental return rate.

The corrected G_{env} ablation (threshold=0.5, matching the dataset median) confirms that the ecology-derived adaptive calibration provides no measurable benefit over a fixed comparator at current estimation quality, isolating G_{env} estimation as the mechanistic bottleneck rather than the MVT switching framework itself.

Limitations. First, our evaluation uses Llama-3.1-8B as the reader. Larger models with stronger instruction-following may tolerate sparse retrieved contexts better, potentially narrowing the F1 gap at 1.3 chunks per question. Second, QASPER’s short, citation-heavy gold answers amplify the penalty for under-retrieval: when the one correct sentence is absent from the retrieved set, F1=0 regardless of answer quality. Third, the multi-hop analysis uses number of questions per paper as a proxy for cross-section evidence requirements; a more principled stratification would require annotating whether gold evidence spans cross section boundaries. Fourth, section parsing relies on header-based heuristics; papers with non-standard structures reduce the accuracy of section detection. Fifth, we do not evaluate top- k -1 or top- k -2 directly, which would provide the tightest efficiency-matched baselines; the Pareto frontier result is conditional on the nine methods tested.

9 Conclusion

We have proposed and evaluated MVT-RAG, the first application of the Marginal Value Theorem from ecological foraging to adaptive section switching in scientific RAG. Treating document sections as foraging patches and switching sections when marginal semantic information gain falls below the document-wide average yields a training-free, LLM-call-free retrieval algorithm that achieves $3.8\times$ fewer chunks per question than top- k -5 and is Pareto-non-dominated among evaluated methods. On QASPER, MVT-RAG significantly outperforms the no-RAG baseline (+0.073 F1, $p<0.001$) but significantly underperforms top- k -5 (−0.079 F1, $p<0.001$). Oracle retrieval analysis localizes the failure to under-retrieval, not generation quality. A corrected ablation study—reconciling a prior threshold specification error to use 0.5 matching the dataset median G_{env} —finds no significant benefit for the ecology-derived adaptive threshold over a fixed comparator. We identify multi-chunk

G_{env} averaging (mean of top- K similarities per section) as the minimal, theoretically grounded fix and provide a mechanistic argument for why it will allow more retrieval per section without sacrificing the efficiency advantage.

Future directions: (1) implement and evaluate the multi-chunk $G_{\text{env}}^{(K)}$ estimator; (2) add section-visit recall as a diagnostic metric to assess whether the section-switching logic correctly identifies gold-evidence sections in multi-hop questions; (3) combine MVT switching with within-section cross-encoder re-ranking to improve oracle recall while preserving the framework’s computational efficiency.

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